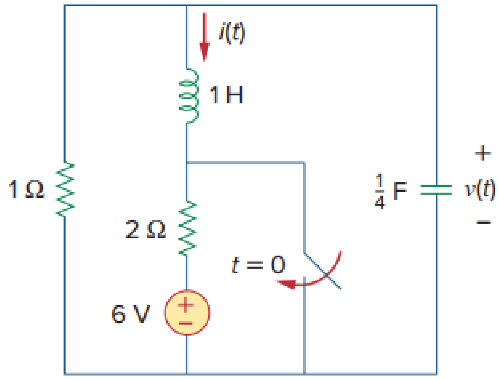
Chapter 8, Problem 48P

Problem

Given the circuit in Fig. 8.95, find $i(t)$ and $v(t)$ for $t > 0$.

Figure 8.95:



Step-by-step solution

Step 1 of 11

The following is the given circuit diagram:

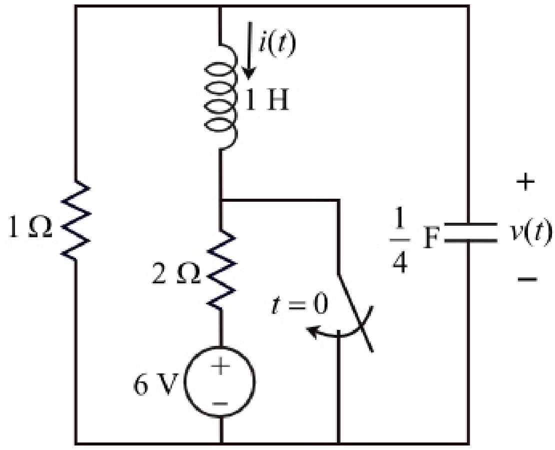


Figure 1

Step 2 of 11

For $t < 0$, the switch is in open position and the circuit is in steady state, therefore the capacitor is replaced by an open circuit and the inductor by a short circuit as shown in Figure 2.

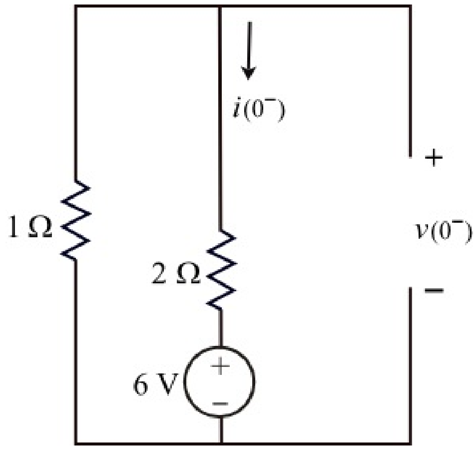


Figure 2

Step 3 of 11

Apply Kirchhoff's voltage law to the left side loop if Figure 2.

$$2i\left(0^-\right)+6+1i\left(0^-\right)=0$$
$$i\left(0^-\right)=\frac{-6}{2+1}$$
$$i\left(0^-\right)=-2\text{ A}$$

Apply Kirchhoff's voltage law to the right side loop if Figure 2.

$$2i\left(0^-\right)+6-v\left(0^-\right)=0$$
$$\begin{aligned}v\left(0^-\right)&=2i\left(0^-\right)+6\\&=2(-2)+6\\&=2\text{ V}\end{aligned}$$

Step 4 of 11

Since the capacitor voltage and inductor current cannot change instantaneously.

$$v\left(0^-\right)=v(0)=2\text{ V}$$
$$i\left(0^-\right)=i(0)=-2\text{ A}$$

Step 5 of 11

For $t > 0$, the switch is closed, we remove the voltage source and the $2\ \Omega$ resistor as they are in parallel with a short circuit. Therefore the equivalent circuit is shown in Figure 3.

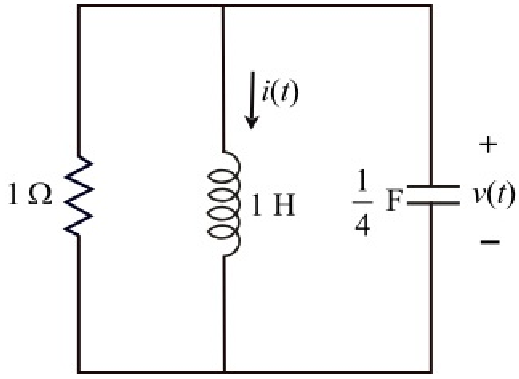


Figure 3

Step 6 of 11

It is a parallel *RLC* circuit. Therefore, the neper frequency is,

$$\begin{aligned}\alpha &= \frac{1}{2RC} \\ &= \frac{1}{2(1)\left(\frac{1}{4}\right)} \\ &= 2 \text{ rad/s}\end{aligned}$$

The resonant frequency is,

$$\begin{aligned}\omega_o &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{(1)\left(\frac{1}{4}\right)}} \\ &= 2 \text{ rad/s}\end{aligned}$$

Step 7 of 11

Since $\alpha = \omega_o$, the response is critically damped and the roots are,

$$\begin{aligned}s_{1,2} &= -\alpha \pm \sqrt{\alpha^2 - \omega_o^2} \\ &= -2 \pm \sqrt{2^2 - 2^2} \\ &= -2\end{aligned}$$

Therefore, the voltage across the capacitor is,

$$v(t) = (A_1 + A_2 t)e^{-\alpha t}$$

$$v(t) = (A_1 + A_2 t)e^{-2t} \quad \ldots\ldots (1)$$

Applying initial conditions, at $t = 0$

$$\begin{aligned}v(0) &= (A_1 + A_2(0))e^{-2(0)} \\ &= A_1\end{aligned}$$

Substitute $v(0)$ in the above equation.

Therefore, $A_1 = 2$

Step 8 of 11

Differentiate the equation (1) with respect to t.

$$\frac{dv(t)}{dt} = -2e^{-2t}\left(A_1 + A_2 t\right) + e^{-2t}A_2$$

At $t = 0^+$,

$$\frac{dv(0^+)}{dt} = -2e^{-2(0)}\left(A_1 + A_2(0)\right) + e^{-2(0)}A_2$$
$$= -2A_1 + A_2$$
$$= -2(2) + A_2$$

$$\frac{dv(0^+)}{dt} = -4 + A_2 \quad \text{...... (2)}$$

Step 9 of 11

Apply Kirchhoff's current law at top node of Figure 3.

$$\frac{v(t)}{1} + i(t) + \frac{1}{4}\frac{dv(t)}{dt} = 0 \quad \text{...... (3)}$$

At $t = 0^+$,

$$\frac{v(0)}{1} + i(0) + \frac{1}{4}\frac{dv(0)}{dt} = 0$$
$$2 + (-2) + \frac{1}{4}\frac{dv(0)}{dt} = 0$$
$$\frac{1}{4}\frac{dv(0)}{dt} = 0$$
$$\frac{dv(0)}{dt} = 0$$

Step 10 of 11

Substitute $\frac{dv(0)}{dt} = 0$ in equation (2).

$$0 = -4 + A_2$$
$$A_2 = 4$$

Substitute $A_1 = 2$ and $A_2 = 4$ in equation (1).

$$v(t) = (2 + 4t)e^{-2t}$$
$$= 2(1 + 2t)e^{-2t}\ \text{V}$$

Therefore, the voltage across the capacitor, $v(t)$ for $t > 0$ is $\boxed{2(1+2t)e^{-2t}\ \text{V}}$.

Step 11 of 11

From equation (3), the current through inductor is,

$$
\begin{aligned}
i(t) &= -\left(v(t)+\frac{1}{4}\frac{dv(t)}{dt}\right) \\
&= -\left((2+4t)e^{-2t}+\frac{1}{4}\frac{d\left((2+4t)e^{-2t}\right)}{dt}\right) \\
&= -\left((2+4t)e^{-2t}+\frac{1}{4}\left[(-2)(2)e^{-2t}+4\left(e^{-2t}+t(-2)e^{-2t}\right)\right]\right) \\
&= -\left((2+4t)e^{-2t}+\frac{1}{4}\left[-4e^{-2t}+\left(4e^{-2t}-8te^{-2t}\right)\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
&= -\left((2+4t)e^{-2t}+\frac{1}{4}\left[-8te^{-2t}\right]\right) \\
&= -\left((2+4t)e^{-2t}-2te^{-2t}\right) \\
&= (-2-2t)e^{-2t}\ \text{A}
\end{aligned}
$$

Therefore, the current through the inductor, $i(t)$ for $t>0$ is $\boxed{(-2-2t)e^{-2t}\ \text{A}}$.